CPSC-406 Report

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Abstract

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1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

1. Word acceptance table

w	accepted by A_1?	accepted by A_2?
aaa	No	Yes
aab	Yes	No
aba	No	No
abb	No	No
baa	No	Yes
bab	No	No
bba	No	No
bbb	No	No

2. Description of the languages

Language of A_1. In A_1, any word beginning with b immediately goes to a trap state. If the word begins with a, the automaton stays in state 2 while reading a's, and moves to the accepting state 3 upon reading a single b. Any further symbol leaves the accepting state.

Therefore, a word is accepted if and only if it consists of one or more a's followed by exactly one b.

$$L(A_1) = \{a^n b \mid n \geq 1\}$$

Equivalently, as a regular expression:

$$L(A_1) = a^+ b$$

Language of A_2. In A_2, the only accepting state is 3. The automaton reaches state 3 precisely after reading two consecutive a's. Any b resets the automaton to state 1.

Thus, a word is accepted if and only if it ends with at least two consecutive a's.

$$L(A_2) = \{w \in \{a, b\}^* \mid w \text{ ends with } aa\}$$

Equivalently, as a regular expression:

$$L(A_2) = (a|b)^* aa$$

Exercise 2

Alphabet: $\Sigma = \{a, b\}$.

1. DFAs for the given languages

(1) All words that end with `ab`. Let $M_1 = (Q, \Sigma, \delta, q_0, F)$ where

$$Q = \{q_0, q_1, q_2\}, \quad q_0 \text{ start}, \quad F = \{q_2\}.$$

Transitions:

	a	b
q_0	q_1	q_0
q_1	q_1	q_2
q_2	q_1	q_0

State q_2 represents that the word currently ends in `ab`.

(2) All words that contain `aba`. Let $M_2 = (Q, \Sigma, \delta, q_0, F)$ where

$$Q = \{q_0, q_1, q_2, q_3\}, \quad q_0 \text{ start}, \quad F = \{q_3\}.$$

Transitions:

	a	b
q_0	q_1	q_0
q_1	q_1	q_2
q_2	q_3	q_0
q_3	q_3	q_3

State q_3 means the substring `aba` has been seen.

(3) Odd number of `a`'s and odd number of `b`'s. Use four states representing parity:

$$Q = \{(E, E), (E, O), (O, E), (O, O)\}$$

(start state (E, E)). Accepting state: (O, O).

Reading a toggles the first component; reading b toggles the second component.

(4) Even number of `a`'s and odd number of `b`'s. Same construction as (3).

Start state: (E, E). Accepting state: (E, O).

Transitions again toggle parity accordingly.

(5) Any three consecutive characters contain at least one `a`. This is equivalent to saying the word does *not* contain `bbb`.

Use states counting consecutive b's:

$$Q = \{q_0, q_1, q_2, q_3\}$$

- q_0: no recent b
- q_1: one consecutive b
- q_2: two consecutive b's
- q_3: three consecutive b's (trap)

Start: q_0.

Accepting states: $\{q_0, q_1, q_2\}$.

Transitions:

- On a: go to q_0 from any state except q_3 - On b:

$$q_0 \to q_1, \quad q_1 \to q_2, \quad q_2 \to q_3, \quad q_3 \to q_3$$

(6) All words that contain `bbb`. Same structure as (5), but now:

Accepting state: $\{q_3\}$ only.

Once q_3 is reached, stay there.

2. Observations

Several patterns appear:

- Languages involving substrings (e.g. `aba`, `bbb`) use states that track partial progress toward matching the substring.
- Parity conditions use a product construction (Cartesian product) of two 2-state automata, giving 4 states.
- Conditions about avoiding a pattern (problem 5) are closely related to conditions about containing that pattern (problem 6); they differ mainly in which states are accepting.
- Many DFAs follow a systematic design pattern: track exactly the minimal information necessary to decide acceptance.

Week 1 Discord Question

Why is it important that the transition function of a DFA be total (defined for every state-symbol pair)? What would break, both formally and conceptually, if we allowed "missing" transitions?

2.2 Week 2

Exercise 1: Product Automata

Let $\Sigma = \{a, b\}$.

1. Description of $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$

Language of $\mathcal{A}^{(1)}$

In $\mathcal{A}^{(1)}$, states 2 and 4 are accepting, and state 3 is a trap state with self-loops on both a and b. From the transition structure we observe:

- From state 2, reading a leads to the trap state.
- From state 4, reading b leads to the trap state.

Thus any occurrence of the substring aa or bb forces the automaton into the trap state.

Therefore,

$$L(\mathcal{A}^{(1)}) = \{w \in \{a, b\}^* \mid w \text{ contains no } aa \text{ and no } bb\}.$$

Equivalently, $L(\mathcal{A}^{(1)})$ consists of all strings whose symbols strictly alternate.

Language of $\mathcal{A}^{(2)}$

In $\mathcal{A}^{(2)}$, state 2 is the only accepting state and state 3 is a trap.

- From the start state 1, reading b leads immediately to the trap.
- Thus every accepted string must begin with a.
- The automaton alternates between states 1 and 2 on every input symbol.

Since state 2 is accepting, the automaton accepts precisely those strings that:

- start with a, and
- have odd length.

Hence,

$$L(\mathcal{A}^{(2)}) = \{w \in \{a,b\}^* \mid w \text{ starts with } a \text{ and } |w| \text{ is odd}\}.$$

2. Construction of the Intersection Automaton

The product automaton has state set

$$Q = Q^{(1)} \times Q^{(2)} = \{1,2,3,4\} \times \{1,2,3\}, \quad \text{start state } (1,1),$$

with transition function defined by

$$\delta\big((i,j),x\big) = \big(\delta_1(i,x), \delta_2(j,x)\big).$$

Reachable states from $(1,1)$ and their transitions:

State	on a	on b
$(1,1)$	$(2,2)$	$(4,3)$
$(2,2)$	$(3,1)$	$(4,1)$
$(4,3)$	$(3,3)$	$(2,3)$
$(3,1)$	$(3,2)$	$(3,3)$
$(4,1)$	$(3,2)$	$(2,3)$
$(3,3)$	$(3,3)$	$(3,3)$
$(2,3)$	$(3,3)$	$(4,3)$
$(3,2)$	$(3,1)$	$(3,1)$

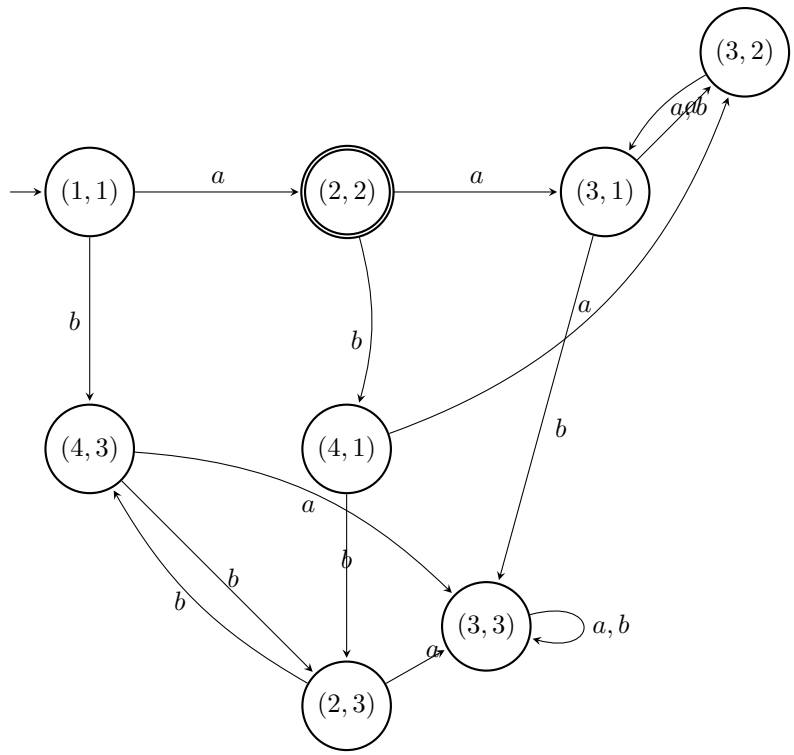
Thus, the reachable states are

$$\{(1,1),(2,2),(4,3),(3,1),(4,1),(3,3),(2,3),(3,2)\}.$$

The accepting states of the intersection automaton are

$$F = F^{(1)} \times F^{(2)} = \{2,4\} \times \{2\} = \{(2,2),(4,2)\}.$$

Since $(4,2)$ is unreachable, the only accepting state is $(2,2)$.



3. Correctness of the Product Construction

By definition of the product construction, a string w is accepted by $\mathcal{A}$ if and only if:

- the run of w in $\mathcal{A}^{(1)}$ ends in an accepting state, and
- the run of w in $\mathcal{A}^{(2)}$ ends in an accepting state.

Thus,

$$L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)}).$$

4. Construction for the Union

To obtain an automaton $\mathcal{A}'$ such that

$$L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)}),$$

we keep:

- the same state set $Q_1 \times Q_2$,
- the same start state $(1, 1)$,
- the same transition function.

We change only the accepting states to:

$$F' = (F_1 \times Q_2) \cup (Q_1 \times F_2).$$

That is, a state (p, q) is accepting if either

$$p \in F_1 \quad \text{or} \quad q \in F_2.$$

Exercise 2: More Automata

Let $\Sigma = \{a, b\}$.

1. Description of the Languages

Language of $\mathcal{B}^{(1)}$

The automaton $\mathcal{B}^{(1)}$ has states $\{p_0, p_1, p_2\}$ where p_0 is the start and only accepting state. Each b leaves the state unchanged, while each a moves cyclically:

$$p_0 \xrightarrow{a} p_1 \xrightarrow{a} p_2 \xrightarrow{a} p_0.$$

Thus the automaton counts the number of a's modulo 3. Since only p_0 is accepting, we obtain

$$L(\mathcal{B}^{(1)}) = \{\, w \in \{a, b\}^* \mid \#_a(w) \equiv 0 \pmod 3 \,\}.$$

In words: all strings whose number of a's is divisible by 3.

Language of $\mathcal{B}^{(2)}$

The automaton $\mathcal{B}^{(2)}$ has states $\{q_0, q_1, q_2\}$ where q_0 is the start and only accepting state, and q_2 is a trap state with loops on both a and b.

The transitions show:

$$q_0 \xrightarrow{a} q_1, \qquad q_1 \xrightarrow{a} q_2.$$

Thus any occurrence of two consecutive a's sends the automaton to the trap state q_2. Therefore,

$$L(\mathcal{B}^{(2)}) = \{\, w \in \{a, b\}^* \mid w \text{ does not contain } aa \,\}.$$

2. Construction of the Intersection Automaton

Define the state set as the Cartesian product:

$$Q = \{p_0, p_1, p_2\} \times \{q_0, q_1, q_2\}, \quad \text{with start state } (p_0, q_0).$$

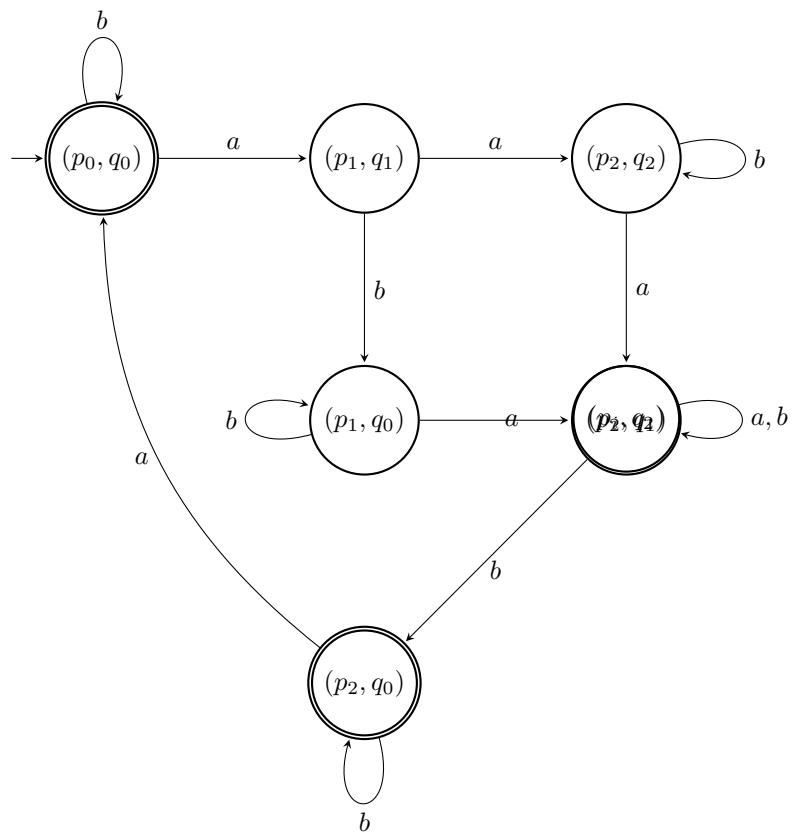
We compute the reachable states and transitions:

State	input a	input b
(p_0, q_0)	(p_1, q_1)	(p_0, q_0)
(p_1, q_1)	(p_2, q_2)	(p_1, q_0)
(p_2, q_2)	(p_0, q_2)	(p_2, q_2)
(p_1, q_0)	(p_2, q_1)	(p_1, q_0)
(p_0, q_2)	(p_1, q_2)	(p_0, q_2)
(p_2, q_1)	(p_0, q_2)	(p_2, q_0)
(p_1, q_2)	(p_2, q_2)	(p_1, q_2)
(p_2, q_0)	(p_0, q_1)	(p_2, q_0)
(p_0, q_1)	(p_1, q_2)	(p_0, q_0)

The accepting states of the intersection automaton are the product of the accepting states of the original DFAs:

$$F = F^{(1)} \times F^{(2)} = \{p_0, p_2\} \times \{q_0\} = \{(p_0, q_0), (p_2, q_0)\}.$$

Among the reachable states, only (p_0, q_0) and (p_2, q_0) satisfy this condition.



Note: The states (p_0, q_2), (p_1, q_2), and (p_2, q_2) are merged into a single trap state for simplicity.

Part 3: Verification. By construction, a string w is accepted by the intersection automaton $\mathcal{B}$ if and only if it is accepted by both original automata, i.e.,

$$w \in L(\mathcal{B}) \iff w \in L(\mathcal{B}^{(1)}) \text{ and } w \in L(\mathcal{B}^{(2)}).$$

Hence, the language recognized by $\mathcal{B}$ is

$$L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)}) = \{w \mid \#_a(w) \not\equiv 1 \pmod 3\} \cap \{w \mid w \text{ contains no } aa\}.$$

3. Correctness of the Construction

By definition of the product construction, a string w is accepted by $\mathcal{B}$ if and only if:

- the run of w in $\mathcal{B}^{(1)}$ ends in p_0, and
- the run of w in $\mathcal{B}^{(2)}$ ends in q_0.

Hence,

$$L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)}).$$

4. Construction of $\mathcal{B}'$ for the Union with a Complement

We want an automaton $\mathcal{B}'$ such that

$$L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}.$$

Step 1: Complement $\mathcal{B}^{(2)}$. Since $\mathcal{B}^{(2)}$ is a complete DFA, we obtain $\overline{\mathcal{B}^{(2)}}$ by swapping accepting and non-accepting states. Thus q_1 and q_2 become accepting, and q_0 becomes rejecting.

Step 2: Product Construction for the Union. We keep:

- the same state set $Q_1 \times Q_2$,
- the same start state (p_0, q_0),
- the same transition function.

We change the accepting states to:

$$F' = (F_1 \times Q_2) \cup (Q_1 \times F_{\overline{2}}).$$

That is, a state (p, q) is accepting if either

$$p = p_0 \quad \text{or} \quad q \in \{q_1, q_2\}.$$

Exercise 2.2.7 — State with Self-Loops

Problem. Consider a DFA $\mathcal{A}$ and a state q such that for every input symbol $a \in \Sigma$, the transition function satisfies $\delta(q, a) = q$. Prove, using induction on the length of input strings, that for any string $w \in \Sigma^*$, the extended transition function satisfies $\hat{\delta}(q, w) = q$.

Solution. We proceed by induction on the length of the string w, denoted $|w|$.

Base case: $|w| = 0$, i.e., $w = \varepsilon$ (the empty string).

By the definition of the extended transition function:

$$\hat{\delta}(q, \varepsilon) = q.$$

Hence, the statement is true for the base case.

Inductive step:

Inductive hypothesis (IH): Assume that for all strings w' of length n, we have $\hat{\delta}(q, w') = q$.

Goal: Show that for any string w of length $n + 1$, $\hat{\delta}(q, w) = q$.

Let w be a string of length $n + 1$, and write $w = w'a$ where $|w'| = n$ and $a \in \Sigma$.

By the recursive definition of $\hat{\delta}$:

$$\hat{\delta}(q, w) = \hat{\delta}(q, w'a) = \delta\big(\hat{\delta}(q, w'), a\big).$$

Applying the inductive hypothesis, $\hat{\delta}(q, w') = q$, so:

$$\delta\big(\hat{\delta}(q, w'), a\big) = \delta(q, a).$$

Since q has a self-loop for every symbol, $\delta(q, a) = q$ for all $a \in \Sigma$. Therefore:

$$\hat{\delta}(q, w) = q.$$

This completes the inductive step, and by induction, the result holds for all $w \in \Sigma^*$.

Week 2 Discord Question

In both exercises, we constructed product automata to recognize intersections and modified acceptance conditions to recognize unions or complements. Suppose instead we were given only one of the product automata (without being told it was constructed from two smaller DFAs). How could we determine whether this automaton can be decomposed into a product of two smaller DFAs?

2.3 Week 3

Homework 1

Let $\Sigma = \{a, b, c\}$ and consider the NFA $\mathcal{A}$.

From the diagram we read:

$$Q = \{0, 1, 2, 3\}, \quad q_0 = 0, \quad F = \{2, 3\}.$$

Transitions:

$$0 \xrightarrow{a,b} 0, \quad 0 \xrightarrow{a} 1, \quad 1 \xrightarrow{b} 2, \quad 1 \xrightarrow{c} 3.$$

1. Description of $L(\mathcal{A})$

From state 0 we may read any number of a's and b's. At some point we must take the transition $0 \xrightarrow{a} 1$, and from state 1 we must finish with either b or c.

Hence the language consists of all strings over $\{a, b\}$, followed by an a, and ending with either b or c.

$$L(\mathcal{A}) = (a|b)^* a(b|c).$$

2. Formal Definition of the NFA

$$\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$$

where

$$Q = \{0, 1, 2, 3\}, \quad \Sigma = \{a, b, c\}, \quad q_0 = 0, \quad F = \{2, 3\}.$$

The transition function δ is:

$$\delta(0, a) = \{0, 1\}, \qquad \delta(0, b) = \{0\}, \quad \delta(0, c) = \varnothing,$$

$$\delta(1, a) = \varnothing, \qquad \delta(1, b) = \{2\}, \quad \delta(1, c) = \{3\},$$

$$\delta(2, x) = \varnothing \text{ for all } x \in \Sigma,$$
$$\delta(3, x) = \varnothing \text{ for all } x \in \Sigma.$$

3. All Paths for $v = abab$

We compute all runs that consume the entire word.

$$0 \xrightarrow{a} \{0,1\}$$

Branch 1:

$$0 \xrightarrow{a} 0 \xrightarrow{b} 0 \xrightarrow{a} 1 \xrightarrow{b} 2$$

This path ends in accepting state 2.

Branch 2:

$$0 \xrightarrow{a} 1 \xrightarrow{b} 2$$

State 2 has no outgoing transitions, so the computation gets stuck before reading the whole word. So it is discarded.

There is exactly one accepting path:

$$0 \xrightarrow{a} 0 \xrightarrow{b} 0 \xrightarrow{a} 1 \xrightarrow{b} 2.$$

4. Determinization via Power Set Construction

Start state:

$$\{0\}.$$

We compute reachable subsets.

$$\begin{aligned}
\delta(\{0\}, a) &= \{0,1\}, \\
\delta(\{0\}, b) &= \{0\}, \\
\delta(\{0\}, c) &= \varnothing.
\end{aligned}$$

$$\begin{aligned}
\delta(\{0,1\}, a) &= \{0,1\}, \\
\delta(\{0,1\}, b) &= \{0,2\}, \\
\delta(\{0,1\}, c) &= \{3\}.
\end{aligned}$$

$$\begin{aligned}
\delta(\{0,2\}, a) &= \{0,1\}, \\
\delta(\{0,2\}, b) &= \{0\}, \\
\delta(\{0,2\}, c) &= \varnothing.
\end{aligned}$$

$$\delta(\{3\}, x) = \varnothing \text{ for all } x \in \Sigma.$$

$$\delta(\varnothing, x) = \varnothing \text{ for all } x \in \Sigma.$$

Reachable DFA states:

$$\{0\}, \quad \{0,1\}, \quad \{0,2\}, \quad \{3\}, \quad \varnothing.$$

Accepting DFA states:

$$\{0,2\}, \quad \{3\}.$$

Transition Table

State	a	b	c
$\{0\}$	$\{0,1\}$	$\{0\}$	$\varnothing$
$\{0,1\}$	$\{0,1\}$	$\{0,2\}$	$\{3\}$
$\{0,2\}$	$\{0,1\}$	$\{0\}$	$\varnothing$
$\{3\}$	$\varnothing$	$\varnothing$	$\varnothing$
$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$

Homework 2

Let $\Sigma = \{0,1\}$ and consider the NFA $\mathcal{A}$ with

$$Q = \{q_0, q_1, q_2, q_3\}, \quad q_0 \text{ start}, \quad F = \{q_3\}.$$

The transition function δ is:

$$\begin{aligned}
&\delta(q_0,0) = \{q_0\}, && \delta(q_0,1) = \{q_0,q_1\},\\
&\delta(q_1,0) = \varnothing, && \delta(q_1,1) = \{q_2\},\\
&\delta(q_2,0) = \{q_1,q_3\}, && \delta(q_2,1) = \varnothing,\\
&\delta(q_3,0) = \{q_3\}, && \delta(q_3,1) = \{q_3\}.
\end{aligned}$$

1. Determinization via the Power Set Construction

The start state of the DFA is:

$$\{q_0\}.$$

We compute all reachable subsets.

Reachable DFA States

$$\begin{aligned}
S_0 &= \{q_0\}\\
S_1 &= \{q_0,q_1\}\\
S_2 &= \{q_0,q_1,q_2\}\\
S_3 &= \{q_0,q_1,q_3\}\\
S_4 &= \{q_0,q_3\}\\
S_5 &= \{q_0,q_1,q_2,q_3\}
\end{aligned}$$

Accepting states are those containing q_3:

$$S_3,\ S_4,\ S_5.$$

Transition Table of $\mathcal{A}^{\mathrm{D}}$

State	0	1
$\{q_0\}$	$\{q_0\}$	$\{q_0, q_1\}$
$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0, q_1, q_2\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_1, q_2\}$
$\{q_0, q_1, q_3\}$	$\{q_0, q_3\}$	$\{q_0, q_1, q_2, q_3\}$
$\{q_0, q_3\}$	$\{q_0, q_3\}$	$\{q_0, q_1, q_3\}$
$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_1, q_2, q_3\}$

Graphical Description

Rename the DFA states:

$$\begin{aligned}
A &= \{q_0\} \\
B &= \{q_0, q_1\} \\
C &= \{q_0, q_1, q_2\} \\
D &= \{q_0, q_1, q_3\} \quad \text{(accepting)} \\
E &= \{q_0, q_3\} \quad \text{(accepting)} \\
F &= \{q_0, q_1, q_2, q_3\} \quad \text{(accepting)}
\end{aligned}$$

Key transitions:

$$\begin{aligned}
A &\xrightarrow{1} B, \\
B &\xrightarrow{1} C, \\
C &\xrightarrow{0} D, \\
D &\xrightarrow{0} E, \\
E &\xrightarrow{1} D.
\end{aligned}$$

All accepting states remain accepting under further input.

2. Is There a Smaller DFA?

Yes.

Observe that all states containing q_3 are equivalent, because:

- Once q_3 is reached, the automaton remains in states containing q_3.
- All such states accept every continuation.

So the states

$$\{q_0, q_1, q_3\}, \quad \{q_0, q_3\}, \quad \{q_0, q_1, q_2, q_3\}$$

are equivalent and can be merged.

After minimization, the DFA has **4 states**:

1. Start state
2. After reading one 1

3. After reading 11
4. Accepting sink state (once 110 appears)

There exists a smaller DFA with **4 states** accepting the same language.

Week 3 Discord Question

Is there a structural way to predict which subsets will collapse during minimization before fully constructing the DFA? In other words, can we detect equivalent subset-states early from properties of the original NFA?

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, Title, Publisher, Year.